

1.530 Shane thinks of three integers. Added two at a time, their sums are 37, 41, and 44. What is the product of the three integers?

1.340 Keith wants to mark all of the original prices in his store down to half-off. The prices are already marked down by 20%. What additional percent should be discounted? Express your answer as a percent rounded to the tenth.

1.440 On a long hike, Greg forgets to fill his canteen. Fortunately, Harold has 40 ounces of water and Jake has 60 ounces. The three hikers share the water so that each gets an equal amount. At the end of the hike, Greg gives Harold and Jake \$5 to split for the water they gave him. How much should Harold get?

1.640 Rebecca runs a quarter-mile each week. In each of the last 6 weeks her time has improved (decreased) by 2%, and she can now run a quarter-mile in 66 seconds. What was her time six weeks ago? Round your decimal answer to the tenth of a second.

1.550 Franklin rides his bicycle to the beach averaging 18mph. He rides home along the same route. What speed must he average on the way home if he wants to average 21 miles per hour for the round trip?

1.650 Congruent circles of radius 6 have their centers at $C(5, 4)$ and $D(1, 2)$. The circles intersect at points A and B. What is the slope-intercept form equation of the line which passes through A and B?

1.121 The perimeter of a rectangle is 16cm, and the area of the same rectangle is 8cm^2 . What is the diagonal length of the rectangle? (See section 5.3).

1.741 The surfaces of a rectangular prism have areas of 12cm^2 , 16cm^2 , and 27cm^2 . What is the volume of the prism?

1.251 The mean of a set of seven positive integers is 7, the median is 8, and the distinct mode is 9. What is the greatest possible range for the set of integers?

1.351 Four integers are added to the set $\{3, 4, 5, 5, 8\}$, increasing the mean, median, and mode each by 1. What is the greatest integer in the new set?

1.551 In a set of ten integers, the average of the two smallest integers is 102, the average of the three smallest integers is 103, the four smallest is 104, and so forth until the average of all ten integers is 110. What is the greatest of these integers?

1.161 The 5th term of an arithmetic sequence is 18 and the 9th term is 24. What is the 99th term?

- 1.661 Find the perimeter of the smallest triangle whose integer side lengths form a geometric sequence whose common ratio r is not equal to 1.
(See p.178).

1.171 Simplify:
$$\frac{5}{6 + \frac{5}{6 + \frac{5}{6 + \dots}}}$$

1.271 Simplify: $20 + \sqrt{20 + \sqrt{20 + \sqrt{20 + \dots}}}$.

1.471 The formula for the sum of the powers of 2 from 2^0 to 2^n is $2^0 + 2^1 + 2^2 \dots + 2^n = 2^{n+1} - 1$. Write a formula which could be used to find the sum of the powers of a^0 through a^n in terms of a and n .

1.571 Simplify: $1 + \frac{3}{2 + \frac{3}{1 + \frac{3}{2 + \frac{3}{1 + \dots}}}}$.

1.081 The product of two numbers is 30 and their sum is 16.
Express their difference in simplest radical form.

1.581 For positive numbers a and b , the positive difference of a and b is divided by the positive difference of their reciprocals and the result is 5. What is the product of a and b ?

1.781 Tejas begins to paint a fence that is 100 meters long at noon. By 2pm, Tejas has one third of the fence painted and Jason arrives to help. At 3pm, four-fifths of the fence is painted and Tejas leaves Jason to finish painting the remaining fence. What time does Jason finish painting the fence?

- 1.391 A solid rubber ball rebounds to three-fourths the height it is dropped from. What is the total vertical distance traveled by a ball dropped from a height of 24 feet as it bounces to a rest (for example, when it hits the floor for the second time it will have traveled $24 + 18 + 18 = 60$ feet).

- 1.491 If $ab = 4$, $bc = 5$, and $ac = 10$, find $a^2 + b^2 + c^2$.

1.921 For positive integers a and b , $(a + \sqrt{b})^2 = 9 + 4\sqrt{5}$,
what is $(a + \sqrt{b})^3$? (See section 1.7).

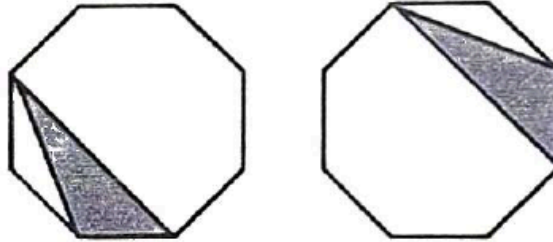
2.210 How many distinct pairs of consecutive integers have a product less than 10,000?

2.120 Ten friends go out to dinner together: 7 order an appetizer, 5 order a soup, and 4 order a salad. If everyone orders something, but no one orders exactly two things, how many people order all three things?

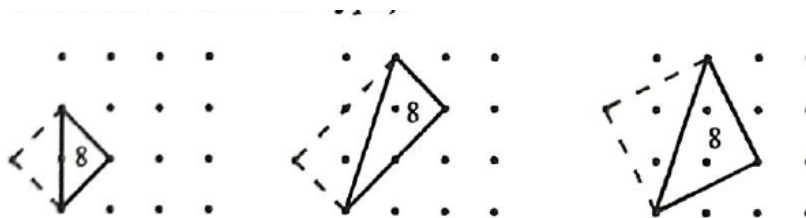
2.320 All squares are both rectangles and rhombuses. All rhombuses and rectangles are parallelograms. On a sheet of paper Josh draws 19 rectangles, 15 rhombuses, and 7 squares. How many parallelograms did Josh draw?

2.520 Students were surveyed on their preference for tea. Three-fourths of the students surveyed like sweet tea, three-fifths of the students surveyed like unsweetened tea, and one sixth of the students do not like either. What is the fewest number of students surveyed who could like both sweet and unsweetened tea?

- 2.640 How many non-congruent triangles can be formed by connecting three of the vertices of a regular octagon. For example, the two triangles drawn below are congruent and should only be counted once.

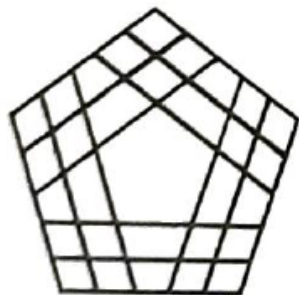


- 2.840 How many right triangles can be formed by connecting three points on the unit grid shown?

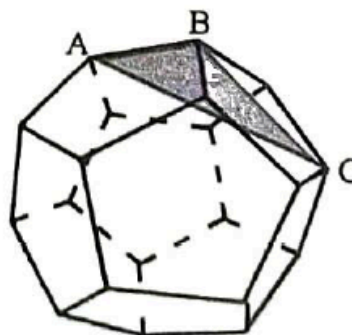


This brings our total to **200**.

- 2.811 How many pentagons can be formed by tracing the lines of the figure below?



- 2.911 A regular dodecahedron has 12 pentagonal faces, 20 vertices, and 30 edges. How many of the triangles which can be formed by connecting three of its vertices have at least one side within the dodecahedron?

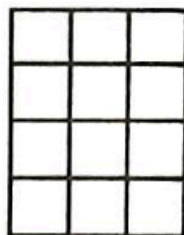


Example: Corey and Tony are friends on the same basketball team. There are eight players on the team. How many starting lineups of five players include Corey, Tony, or both?

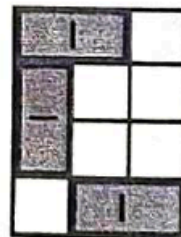
- 2.121 There are seven parking spaces in a row which must be assigned to four co-workers. How many ways can the spaces be assigned if at least two of the assigned spaces must be adjacent?

2.621 Five lines are drawn so that each intersects the other four, but no three lines intersect at the same point. Each of the points of intersection is then connected to every other point of intersection. How many triangles are formed which have three of the points of intersection as vertices?

2.441 Three identical dominoes are placed on the 3×4 tiled board below to create a complete pathway from the upper-left corner to the lower-right corner of the board (no dominoes may touch only at the corners). How many ways can this be done?



Yes



No

- 2.461 Three friends: Al, Bill, and Carl, are painting their chests with the word CAVALIERS for a basketball game, with at least two letters painted on each friend's chest. For example, Al might get "CA", Bill "VAL", and Carl "IERS", or Carl could get "CAVA" while Bill gets "LI" and Al gets "ERS". How many ways can the three friends get their chests painted so that when they stand in the correct order, they spell out the word CAVALIERS?
- 2.561 An elaborate code consists of left and right arrows which are placed together to represent words. There are spaces between words. For example, the code of three left and four right arrows arranged with two spaces: << >>> <> represents a three-word code phrase. How many different three-word code phrases in this system use 8 letters?

2.271 Find $a + b$ if $\binom{9}{5} + \binom{9}{6} = \binom{a}{b}$.

2.081 The 6 slices can be divided among the 4 students who are not vegetarians. Use 3 dividers and 6 slices which gives us $9C3 = 84$ ways to divide the pepperoni slices. The 3 slices of cheese pizza can be divided among all 5 students. Using 4 dividers and 3 slices we get $7C3 = 35$ ways to distribute the cheese slices. This gives us a total of $35 \cdot 84 = \mathbf{2,940}$ ways to distribute all 9 slices.

2.181 How many integers between 1,000,000 and 2,000,000 have at least four fives?

2.381 Three standard six-sided dice are rolled: one red, one green, and one white. How many outcomes are possible where the number rolled on the red die is greater than the number rolled on the green die, and the number on the green die is greater than the number rolled on the white die?

3.500 What is the probability that the top four cards of a shuffled deck are all of the same suit?

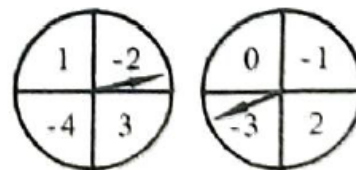
3.700 What is the probability that the top card in a shuffled deck is an Ace (of any color), and the second card is a spade? The Ace may be the Ace of spades.

3.510 A dot is added to each face of two standard dice. What is the probability of rolling an 8 with the modified pair of dice?

3.610 For three randomly selected positive integers, what is the probability that the sum of the units digits is even?

3.810 Each spinner below is divided into four equal sectors.

Each is spun once, and the product of the spins is calculated. What is the probability that the product is positive?



$$P(\text{both negative}) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}.$$

Add these together to get a probability of $\frac{3}{8}$.

Example: Lisa shakes her piggy bank which contains 5 quarters and 5 nickels until 5 coins fall out. Assuming nickels and quarters fall out with equal probability, what is the probability that the sum of the change that has fallen out is \$0.85?

Example: There are a dozen eggs in a basket and four have been hard-boiled. You can tell by spinning an egg whether it has been hard-boiled. You select three eggs from the basket. What is the probability that only one of the three is hard-boiled?

- 3.030 Three numbers are selected at random and without replacement from a bag containing each of the numbers 1 through 9. What is the probability that the three numbers are consecutive integers?